BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type (Group-I)

Time Allowed: 30 Min. | **Maximum Marks:** 20

Note: Inter, Part I, 2022 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. A matrix of order 1 × n is called:

(A) Unit matrix
(B) Diagonal matrix
(C) Column matrix
(D) Row matrix

Q2. The rational fraction x^2/(x^2 − 1) is:

(A) Improper
(B) Proper
(C) Polynomial
(D) Equation

Q3. The 3rd term of the sequence a_n = (−1)^n (n + 7) is:

(A) 8
(B) 4
(C) -8
(D) -4

Q4. [illegible in source scan: question text obscured by watermark]

(A) 0
(B) 1
(C) ω
(D) ω^2

Q5. If |a b; c d| = 2, then |b a; d c| = :

(A) 2
(B) -2
(C) ±2
(D) 0

Q6. In the set of complex numbers C, the multiplicative identity is:

(A) (1, 0)
(B) (0, 1)
(C) (1, 1)
(D) (0, 0)

Q7. Roots of quadratic equation ax^2 + bx + c = 0 are equal if:

(A) b^2 − 4ac < 0
(B) b^2 − 4ac > 0
(C) b^2 − 4ac ≥ 0
(D) b^2 − 4ac = 0

Q8. If a, b are negative and G.M. is also negative, then:

(A) H < A < G
(B) A < H < G
(C) G < A < H
(D) A < G < H

Q9. $^{n}P_n$ = :

(A) n!
(B) n
(C) 0
(D) 1

Q10. Set N of natural numbers is monoid w.r.t.:

(A) +
(B) −
(C) ×
(D) ÷

Q11. If "n" is a positive integer, then n^3 + n is divisible by:

(A) 0
(B) 2
(C) 3
(D) 5

Q12. In right triangle ABC, if β = 90°, then cosine law reduces to:

(A) b^2 = c^2 + a^2
(B) c^2 = a^2 + b^2
(C) a^2 = b^2 + c^2
(D) b^2 = c^2 − a^2

Q13. Which of the following is a quadrant angle:

(A) 350°
(B) 390°
(C) −368°
(D) 410°

Q14. Angle below the horizontal line is called:

(A) corresponding angle
(B) Alternate angle
(C) Angle of depression
(D) Angle of elevation

Q15. A die is rolled, then the probability of getting prime number is:

(A) 1/6
(B) 1/4
(C) 1/3
(D) 1/2

Q16. Reference angle of tan x = −1 equals:

(A) π/3 (B) π/4

(C) −π/4 (D) −π/3

Q17. In the expansion (a + x)^n if "n" is even, then middle term is:

(A) ((n+1)/2)th term (B) (n/2 + 1)th term

(C) (n/2)th term (D) ((n+3)/2)th term

Q18. Range of y = sec x is:

(A) [−1, 1] (B) [0, π]

(C) x ≤ −1, x ≥ 1 (D) y ≤ −1, y ≥ 1

Q19. cos(tan^−1√3) = :

(A) √3/2 (B) −√3/2

(C) 1/2 (D) −1/2

Q20. cos 48° + cos 12° = :

(A) √3 cos 18° (B) √2 cos 18°

(C) 2 cos 18° (D) 3 cos 18°

Essay Type (Group-I)

Time Allowed: 2.30 hrs. | **Maximum Marks:** 80

QSECTION – I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Check the closure property w.r.t. "×" on (-1, 1)

ii Simplify the complex numbers (5, -4) (-3, 2)

iii Find the multiplicative inverse of complex number (√2, −√5)

iv Write the descriptive and tabular form of {x / x ∈ P ∧ x < 12}

v Verify the commutative property of union and intersection for sets A = {1,2,3,4,5}, B = {4,6,8,10}

vi Write inverse and contrapositive of the condition ¬p → q

vii Find x and y if [x+3 1; −3 3y−4] = [y 1; −3 2x]

viii If A and B are non-singular matrices, then show that (AB)^−1 = B^−1A^−1

ix Show that |α β+γ 1; β γ+α 1; γ α+β 1| = 0

x Solve the equation x^2 − 7x + 10 = 0 by factorization.

xi Reduce $2x^4 - 3x^3 - x^2 - 3x + 2 = 0$ into quadratic form.

xii Find four fourth roots of unity.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Define conditional equation. Give one example.

ii Write only partial fractional form of $x^2 / ((x^2+1)^2(x-1))$

iii Write first two terms of sequence in which $a_n = (-1)^n n^2$

iv How many terms are in A.P in which $a_1 = 11$; $a_n = 68$; $d = 3$

v Find 11th term of G.P; $1 + i, 2, 2(1 - i)$, ----------

vi If '5' is H.M. between 2 and b, then find b

vii Write formula for $\Sigma^n n$ and $\Sigma^n n^2$

viii Write $(10 \cdot 9)/(2 \cdot 1)$ into factorial form.

ix Show that ${}^nP_r = n \cdot {}^{n-1}P_{r-1}$

x Use mathematical induction to prove that $1 + 3 + 5 + ... + (2n - 1) = n^2$ is true for n = 1 and n = 2.

xi Find middle term in expansion of $(x/2 + 2/x^2)^12$

xii Expand $(1 - x)^{1/2}$ upto 3 terms.

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Find r, when l = 5 cm, $\theta = 1/2$ radians.

ii Verify that $\sin 60° \cos 30° - \cos 60° \sin 30° = \sin 30°$

iii Prove that $\cos^4 \theta - \sin^4 \theta = \cos^2 \theta - \sin^2 \theta$

iv Without using the tables, find the value of $\tan 110°$

v Prove that $\tan(\pi/4 - \theta) + \tan(3\pi/4 + \theta) = 0$

vi Prove that $\cos 20° + \cos 100° + \cos 40° = 0$

vii Find the period of $\operatorname{cosec} x/4$

viii At the top of a cliff 80 m high, the angle of depression of a boat is 12°. How far is the boat from the cliff?

ix Find the measure of the greatest angle, if the sides of a triangle are 16, 20, 33

x Find the area of a triangle ABC, when b = 25.4, $\gamma = 36°41'$, $\alpha = 45°17'$

xi Show that $\sin(2\cos^{-1}x) = 2x\sqrt{(1-x^2)}$

xii Solve sin x + cos x = 0

xiii Find the solution of sin x = √3/2 in (0, 2π)

QSECTION-II. Note: Attempt any THREE questions.

QQ.5(a). (Marks 5) Use matrices to solve: x − 2y + z = 1; 3x + y − 2z = 4; y − z = 1

QQ.5(b). Solve the equation 4^1[illegible in source scan]x + 4^1[illegible in source scan]x = 10

QQ.6(a). Resolve x^4/(1−x^4) into partial fractions.

QQ.6(b). (Marks 5) The sum of three numbers in an A.P is 24 and their product is 440. Find the numbers.

QQ.7(a). Prove that ${}^{n}C_r = {}^{n}C_{r-1} = {}^{n+1}C_r$

QQ.7(b). Use binomial theorem to show that 1 + 1/4 + 1·3/(4·8) + ---- = √2

QQ.8(a). If cosec θ = (m^2+1)/(2m) and m > 0 (0 < θ < π/2), find the values of remaining trigonometric ratios.

QQ.8(b). If α, β, γ are angles of a triangle ABC, show that cot(α/2) + cot(β/2) + cot(γ/2) = cot(α/2)cot(β/2)cot(γ/2)

QQ.9(a). Prove that r_1 + r_2 + r_3 − r = 4R

QQ.9(b). Prove that tan^−1(120/119) = 2cos^−1(12/13)

GROUP – II

Objective Type (Group-II)

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Inter, Part I, 2022 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Let = (3,4) be n complex number, then value of cost? equals:

(A) 3/4

(B) 1/3

(C) 3/5

(D) 4/5

Q2. If a and b are elements of group G, then solution of equation ax = b will be:

(A) a^−1b
(B) b^−1a
(C) ab
(D) ba^−1

Q3. Let A be a square matrix of order 3×3, and |A| = 2, then |2A| is equal to:

(A) 12
(B) 16
(C) 8
(D) 4

Q4. Square matrix 'A' will be called skew symmetric if A^t equals:

(A) A^t
(B) −A^t
(C) A
(D) −A

Q5. Let the roots of equation x^2 − √m x + 1 = 0 be real and equal, then 'm' is equal to:

(A) 3
(B) 4
(C) 2
(D) 1

Q6. If the sum of roots of equation ax^2 + bx + c = 0 is 1, then:

(A) a + c = 0
(B) a − c = 0
(C) a + b = 0
(D) a − b = 0

Q7. The fraction will be proper rational fraction if degree of p(x) is:

(A) 4
(B) 3
(C) 2
(D) 1

Q8. The sum of 10 A.Ms between 2 and 6 will be:

(A) 40
(B) 30
(C) 20
(D) 10

Q9. 7th term of H.P; 1/3, 1/5, 1/7, ---- will be:

(A) 15
(B) 1/15
(C) 21
(D) 1/21

Q10. Total numbers of triangles formed by 5 sided figure equals:

(A) 4
(B) 6
(C) 8
(D) 10

Q11. The probability of getting an 'ace' card from a deck of 52 playing cards is:

(A) 1/13
(B) 2/13
(C) 3/13
(D) 4/13

Q12. Middle term in the expansion of (a + b)^6 will be:

(A) T_6 (B) T_5

(C) T_4 (D) T_3

Q13. If "n" is a positive integer, then n^3 + n is divisible by:

(A) 0 (B) 2

(C) 3 (D) 5

Q14. Area of sector of circular region of radius r and central angle θ will be:

(A) rθ^2 (B) 1/2 rθ^2

(C) 1/2 r^2θ (D) r^2θ

Q15. (1 − cos α)/sin α = :

(A) cot α/2 (B) tan α/2

(C) tan α (D) cot α

Q16. Period of tan 5x is equal to:

(A) π (B) 5π

(C) 2π/5 (D) π/5

Q17. In ΔABC, if Δ = 2; a = 2; γ = 30°, then value of 'b' equals:

(A) 1 (B) 2

(C) 3 (D) 4

Q18. The radius 'r' of inscribed circle is equal to:

(A) Δ/(s−a) (B) Δ/s

(C) Δ/(s−b) (D) Δ/(s−c)

Q19. sin[tan^−1(−1)] equals:

(A) [illegible in source scan] (B) 1

(C) −1/√2 (D) 1/√2

Q20. If cosec θ = 2 and θ ∈ [0,2π], then its solution is:

(A) {π/6, 5π/6} (B) {π/6, 5π/3}

(C) {π/3, 5π/3} (D) {π/3, 5π/6}

Essay Type (Group-II)

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

QSECTION – I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Factorize 9a^2 + 16b^2

ii Prove that overline{z_1 + z_2} = overline{z_1} + overline{z_2} ∀z_1,z_2 ∈ C

iii Show that (z − overline{z})^2 is real number.

iv Show that whether set of rational numbers is a group under multiplication.

v If a, b are elements of a group G, then show that (ab)^−1 = b^−1a^−1

vi Construct truth table of (p → ¬p) ∨ (p → q)

vii Define diagonal matrix. Give example.

viii Solve for matrix X, 3x − 2A = B if A = [2 3 −2; −1 1 5], B = [2 −3 1; 5 4 −1]

ix Find the value of x if |3 1 x; −1 3 4; x 1 0| = −30

x The sum of a positive number and its reciprocal is 26/5. Find number.

xi Solve by quadratic formula (x − b)(x − a) + (x − b)(x − c) + (x − c)(x − a) = 0

xii Prove that x^3 + y^3 = (x + y)(x + ωy)(x + ω^2y)

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Define proper rational fraction.

ii Write partial fractions for (4x^2+8)/(x^4+2x^2+9) without finding the unknown constants.

iii If a_n−3 = 2n − 5, find the nth term of the sequence.

iv If 1/a, 1/b, 1/c are in A.P, show that b = 2ac/(a+c)

v Insert four real geometric means between 3 and 96

vi If 5 is the H.M. between 2 and b, find the value of b

vii Find the 5th term of the G.P. 3, 6, 12, ----------

viii In how many ways can 4 keys be arranged on a circular key ring?

ix Explain the terms sample space and event.

x Prove the formula 1+5+9+ ... ...+ (4n − 3) = n(2n − 1) for n = 1,2

xi By means of binomial theorem, calculate (0.97)^3

xii Expand (4 − 3x)^1/2 upto 4 terms given that expansion is valid.

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Convert 75°6'30" into radian.

ii Verify that 2 sin 45° + 1/2 cosec 45° = 3/√2

iii Prove that cot^2θ − cos^2θ = cot^2θ cos^2θ

iv Prove that (cos8° − sin8°)/(cos8° + sin8°) = tan37°

v Prove that 1 + tan α tan 2α = sec 2α

vi Express cos 7θ − cos θ as product.

vii Find the period of 3 cos x/5

viii In the right triangle ABC, α = 37°20', a = 243, γ = 90°. Find 'β' and c.

ix Find area of ΔABC in which a = 18, b = 24, c = 30

x Show that r_2 = s tan(β/2) (with usual notations)

xi Find the value of cos^−1(−1/2)

xii Find the solutions of equation which lies in [0, 2π]; sec x = 2

xiii Find the solutions of equation sin x = √3/2 which lies in [0,2π]

QSECTION-II. Note: Attempt any THREE questions.

QQ.5(a). Solve the system of linear equations: 2x_1 − x_2 + x_3 = 8; x_1 + 2x_2 + 2x_3 = 6; x_1 − 2x_2 − x_3 = 1 by Cramer's rule.

QQ.5(b). Prove that x^2/a^2 + (mx+c)^2/b^2 = 1 will have equal roots if c^2 = a^2m^2 + b^2; a ≠ 0, b ≠ 0

QQ.6(a). Resolve into partial fraction 1/((x−3)^2(x+1))

QQ.6(b). The sum of an infinite geometric series is 9 and the sum of the squares of its terms is 81/5. Find the series.

QQ.7(a). Find n and r if ${}^{n-1}C_{r-1} : {}^{n}C_{r} : {}^{n+1}C_{r+1} = 3 : 6 : 11$

QQ.7(b). Find the term independent of x in the expansion of (1+x^2)^3(1+1/x^2)^4

QQ.8(a). If cot θ = 15/8 and terminal arm of the angle is not in quadrant 1, find the value of cosθ and cosec θ

QQ.8(b). Reduce sin^4θ to an expression involving only function of multiples of θ, raised to the first power.

QQ.9(a). In any triangle ABC, with usual notation, prove that a^2 = b^2 + c^2 − 2bc cos α

QQ.9(b). Prove that tan^−1(1/11) + tan^−1(5/6) = tan^−1(1/3) + tan^−1(1/2)